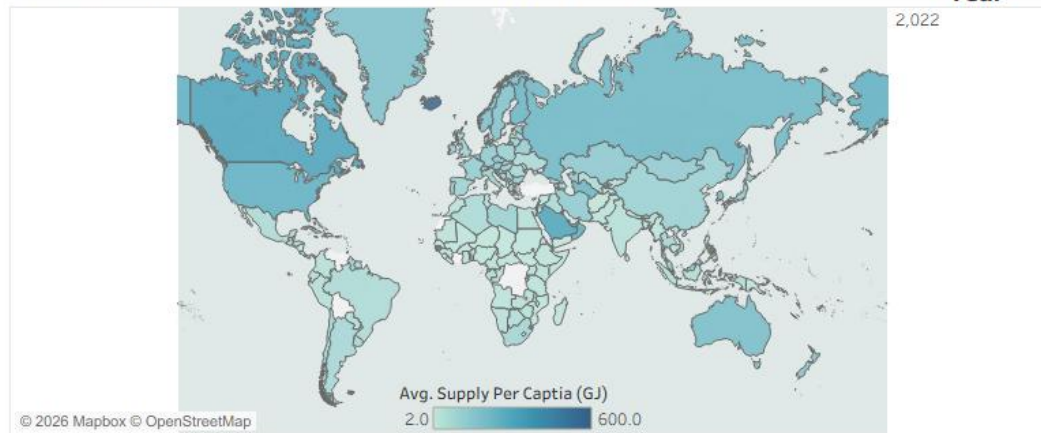
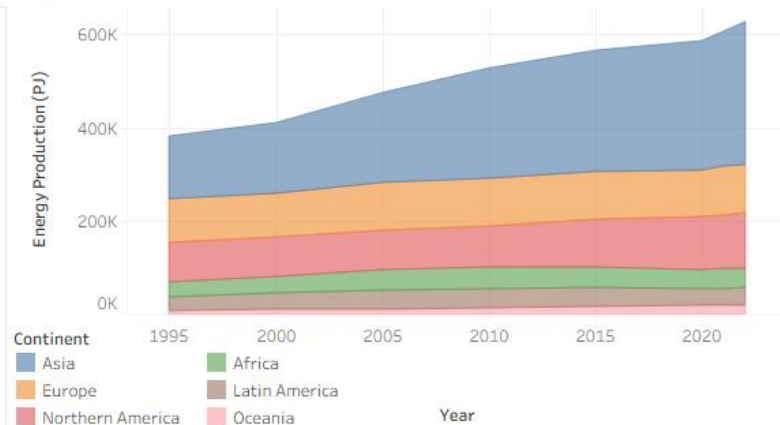


The Global Energy Divide: Production, Trade and Per Capita Supply, 1995–2022

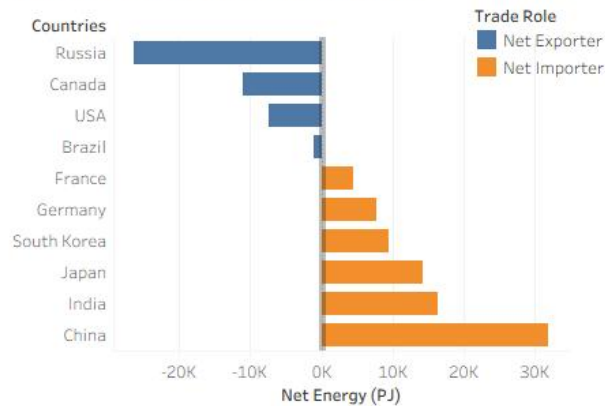
Energy Supply Per Capita(GJ, log scale) by Country



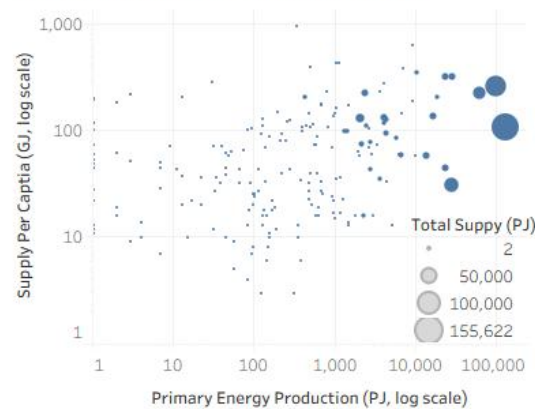
Continental Global Energy Production



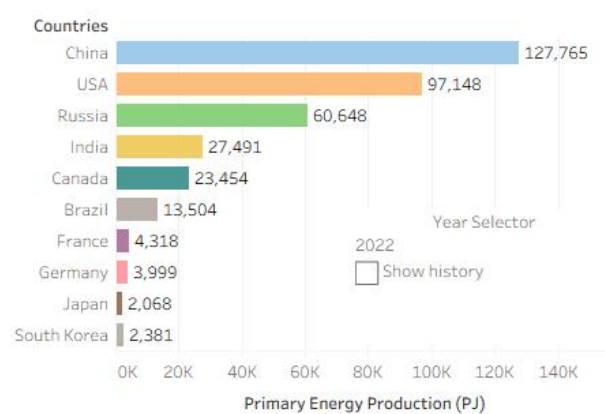
Top 10 Net Importers and Exporters



Production vs Per Capita Supply



Top 10 Energy Producers by Year, 1995–2022



The Global Energy Divide: Production, Trade and Per Capita Supply, 1995 - 2022

This visualisation explores whether nations that produce the most energy necessarily provide the most to their populations, targeting policy analysts and geographically literate readers. The UN Energy Statistics Yearbook 2022 dataset spans five series across 235 countries and eight time points providing multivariate complexity across spatial and temporal dimensions.

The choropleth map encodes supply per capita through a sequential single hue blue palette which is appropriate because the variable is unidirectional with no meaningful midpoint following Harrower and Brewer's (2003) guidance that diverging palettes require a conceptually significant centre value.

The colour scale was clamped at 600 GJ instead of using a logarithmic scale. I chose to clamp the scale at this point because Greenland acted as an extreme outlier as its value was significantly high and it compressed the colour variation between all other countries into an indistinguishable range. By clamping the scale meaningful differences in energy usage across Europe and Asia become visible reflecting Cairo's (2016) principle that honest visualisation requires transparent choices to highlight variation effectively.

The Natural Earth projection was selected following Snyder's (1987) recommendation for world level thematic maps where shape recognition takes precedence over area preservation. An interactive year parameter updates the map, trade balances chart and scatter plot simultaneously constituting a selection interaction in Heer and Shneiderman's (2012) study. A highlight action further enables users to hover over any country to surface its specific data across all components.

Temporal trends are captured through a continental stacked area chart and an animated bar chart. The area chart uses position on a common baseline, which Bertin (2011) identifies as appropriate for ordered quantitative data, encoding absolute global production growth and Asia's rising proportional dominance. Six regions are distinguished by a colourblind safe qualitative palette to ensure accessibility.

The scatter plot is the analytical centrepiece encoding production on a logarithmic x-axis, supply per capita on a logarithmic y-axis and total supply as a marked as circle size. Both axes use logarithmic scales because the data spans several orders of magnitude and a linear scale would make most countries unreadable (Tufte, 2001). The chart exposes the energy paradox directly showing that nations with substantial production values cluster near the bottom of the y-axis, confirming that extraction does not guarantee domestic access. Munzner (2014) notes that size effectively complements position for ordered quantitative data.

The diverging bar chart encodes net trade position using length relative to a zero baseline which is the most perceptually accurate encoding for signed comparison with a colourblind safe two hue palette distinguishing exporters like Russia from importers like India. The animated bar chart uses Tableau's Pages shelf to convey temporal change through movement which highlights China's production growth across 1995 to 2022. The overall Z pattern composition guides the viewer from macro geographic context to granular relational analysis.

A limitation is that country name mismatches between the dataset and shapefile introduced null geometries visible as white polygons on the map. Logarithmic scales are a disadvantage as they require higher graphicacy from the audience to interpret absolute magnitudes correctly (Cairo, 2016).

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